C PROGRAMMING PROJECT

Project Title: Fibonacci Series Generator

File Name: fibonacci.c
Language: C

1. Aim

To write and execute a C program to display the Fibonacci series for a specified number of terms.

2. Algorithm

• Start the program.

• Declare integer variables n, i, a = 0, b = 1, and c.

• Read the number of terms n from the user.

• Repeat n times: print a, calculate c = a + b, then update a = b and b = c.

• Stop the program.

3. C Program

```
#include <stdio.h>

int main()
{
    int n, i, a = 0, b = 1, c;

    printf("Enter the number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for (i = 1; i <= n; i++)
    {
        printf("%d ", a);
        c = a + b;
        a = b;
        b = c;
    }

    return 0;
}
```

4. Sample Output

```
Enter the number of terms: 7
Fibonacci Series: 0 1 1 2 3 5 8
```

5. Result

The C program successfully displays the Fibonacci series for the number of terms entered by the user.

— End of Project —